airtel

SYNC

ENTERPRISE TRACK A

Udyam Digital

Find, buy and activate the right business bundle in one sitting: on a phone, in the owner's language



GenAI advisor | Instant quote | Smart KYC | Live tracker

Watch the demo (2:15)

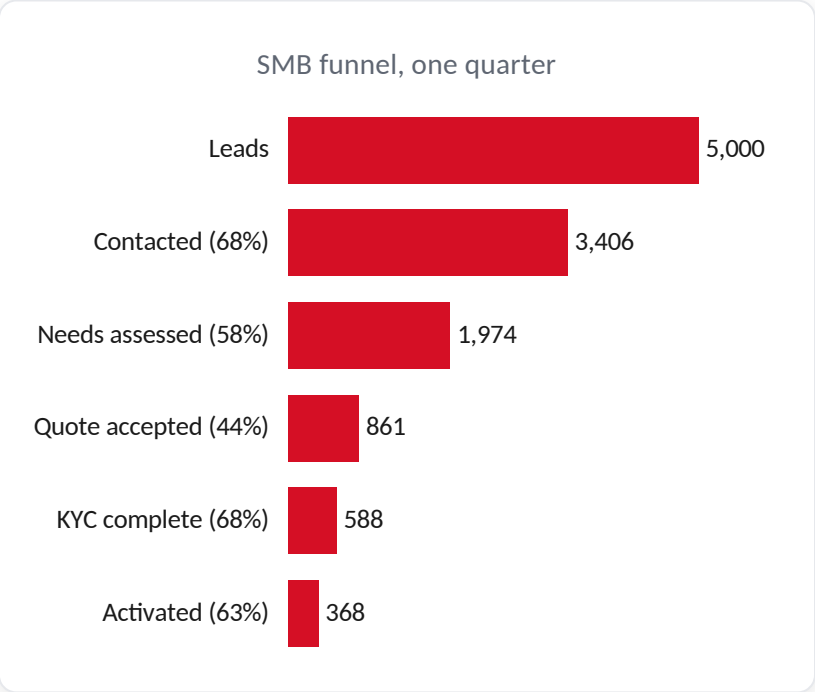
GitHub: ploglabs/udyam-digital

Team hooray · VIT Chennai

Platform Partner unstop

THE PROBLEM

Only 7% of SMB leads activate, and each big drop is fixable



17.6 days
lead to activation

₹15,053
to acquire one SMB

77%
prefer a non-English language

3,026 lost before needs: unreachable, no time for a visit
→ **Owner starts a 24×7 chat in 10 languages**

1,113 lost after needs: price too high, quote too complex
→ **Instant quote, one monthly price**

273 lost at quote: missing documents, long paperwork
→ **Smart KYC with registry prefill**

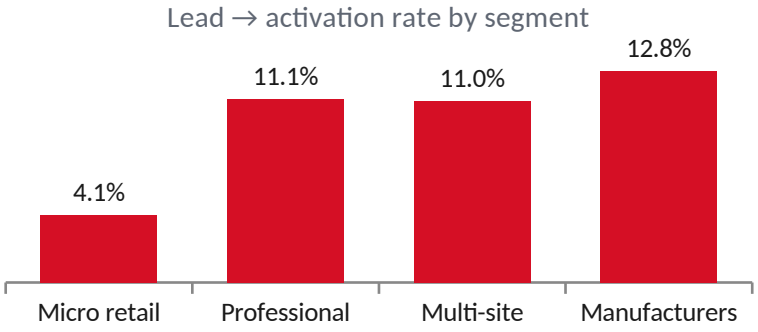
220 lost after KYC: install delays and cancellations
→ **Feasibility upfront, live tracker**

Source: smb_leads.csv, 5,000 leads, Apr–Jun 2026 (synthetic). % = conversion from the previous stage.

Platform Partner unstop

TARGET CUSTOMER AND PRICING

Lead with professional offices; win micro retail with a ₹2.2k digital starter



Segment	Leads	Acq. cost	Payback	Churn 6 mo
Micro retail	54%	₹20,007	8.0 mo	23%
Professional	25%	₹10,695	**1.6 mo**	**14%**
Multi-site	15%	₹15,614	0.5 mo	11%
Manufacturers	6%	₹15,324	0.8 mo	8%

Dukaan Starter

Kirana, salon, pharmacy

₹2,196 /mo

Fibre 100 · 2 SIMs · cloud POS

Self-serve

Clinic & Office Pro

Clinic, CA firm · 8 staff

₹4,592 /mo

Fibre 300 · email · backup · security

Self-serve

Multi-site One

Restaurants, 3 outlets

₹28,045 /mo

SD-WAN · failover · POS · CCTV

Digital + survey

Factory Connect

Small unit · 30 workers

₹22,617 /mo

Leased line · IoT sensors · firewall

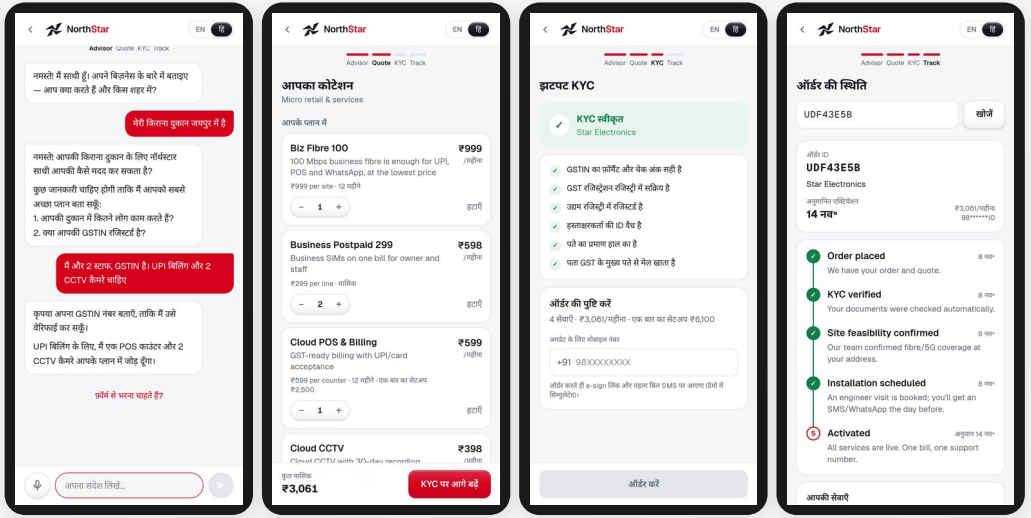
Assisted

Acq. cost = sales and partner spend in segment ÷ activations; payback = acq. cost ÷ avg monthly bundle. Bundle prices: catalogue list prices excl. 18% GST.

Platform Partner unstop

THE SOLUTION

From chat to activation on a phone: no visit, no paperwork



1 Advisor
Chat in Hindi or English

2 Quote
One price, live add-ons

3 Smart KYC
Consent, lookup, doc AI

4 Tracker
Live ETA, WhatsApp alerts

[Watch the full journey: youtu.be/fFOwhewZ7_E](youtu.be/fFOwhewZ7_E)

Smart KYC: 12 applicants, instant verdicts



5 approved

1 re-upload (blurred)

6 rejected

Caught: cancelled GSTIN, tampered check digit, edited legal name, 257-day-old bill, missing Udyam

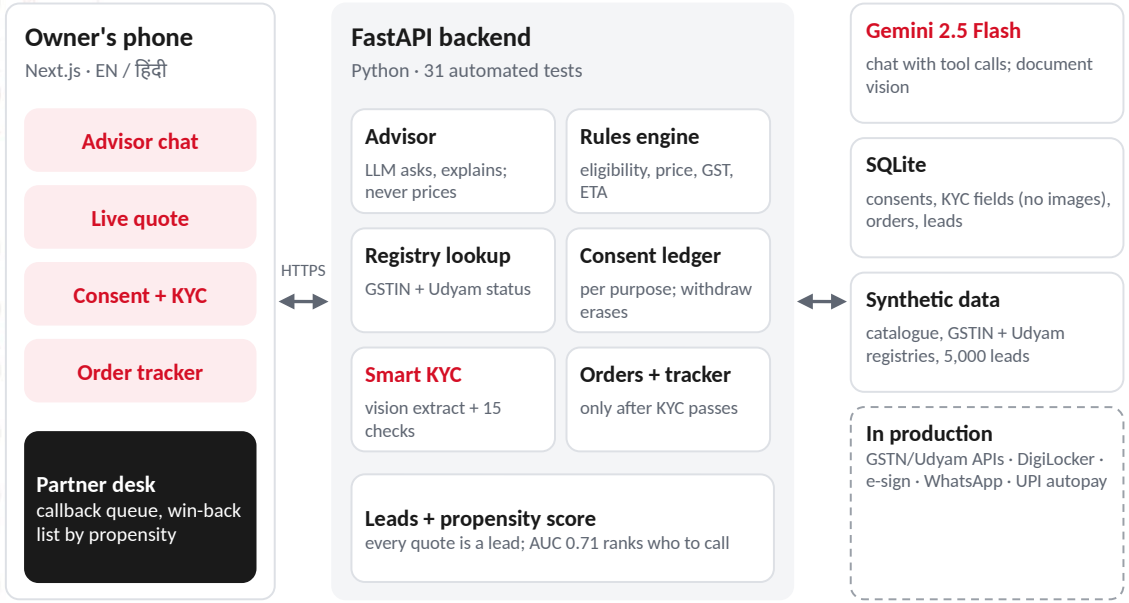
22 customers activated last quarter had a cancelled or suspended GSTIN. A live registry check blocks this at sign-up.

Screens from the working prototype (Next.js + FastAPI + Gemini), shown in Hindi. KYC results from the 12 specimen applicants K01–K12.

Platform Partner unstop

SOLUTION ARCHITECTURE AND DATA PRIVACY

AI explains, rules decide, and personal data stays minimal



PRIVACY AND SECURITY

Designed against the DPDP Act, 2023

- **Consent first** — per purpose, before upload; withdrawal erases data
- **Collect the minimum** — never Aadhaar, PAN, bank details or photos
- **Images never stored** — fields extracted, then the image is dropped
- **Masked everywhere** — GSTIN, Udyam and phone masked in UI, API, logs
- **AI never decides** — verdicts come from rule checks, with reasons
- **Locked-down access** — partner desk behind a key; upload type and size limits

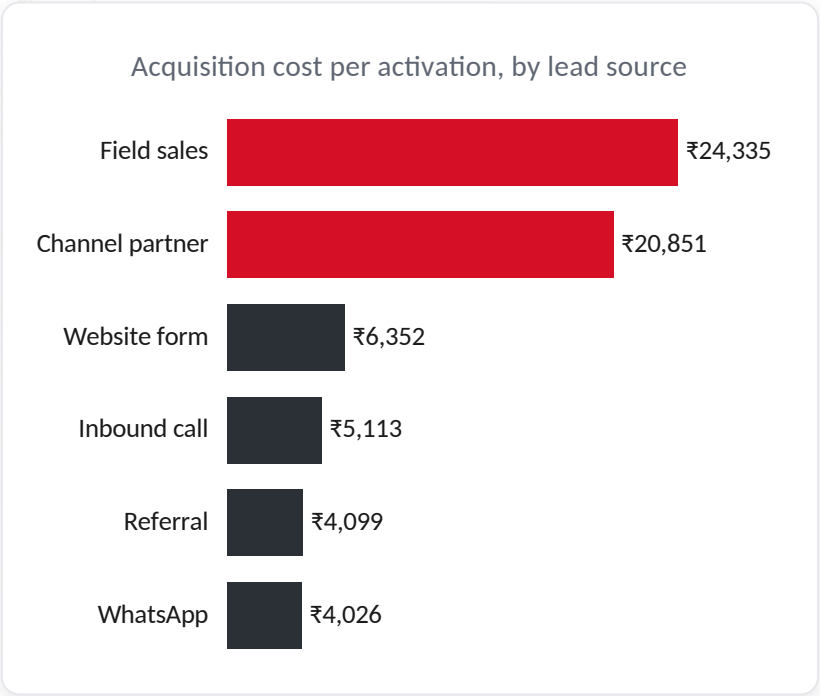
Design rule Gemini talks and reads documents; deterministic code prices, checks and approves. No AI-invented price or verdict.

Solid boxes run in the prototype; dashed box = production integrations. Full diagram and 1-page privacy note: annexure and GitHub (deliverables/).

Platform Partner unstop

GO-TO-MARKET

Digital runs the routine sale; people win the complex accounts



Self-serve digital

Micro retail and professional offices · ~80% of leads

Advisor, quote, KYC and tracker end to end. A person joins only when the owner asks.

Local channel partners

Credited, not bypassed

Partner desk shows the live callback queue and win-back list; commission on QR sign-ups.

Field sales

Multi-site, manufacturers, leased lines

Call list ranked by a propensity score: AUC 0.71, 1.9× lift in the top decile.

Acquisition cost per activation by lead source = spend on leads from that source ÷ activations (smb_leads.csv). Propensity: logistic regression, holdout.

VALUE AND SUCCESS METRICS

Five metrics define success; hitting them doubles activations

	TODAY		TARGET
Lead → activation Self-serve advisor, instant quote	7.4%	→	15%
Days to activate KYC in minutes, feasibility upfront	17.6	→	5
Acquisition cost Digital and partner-referral mix	₹15,053	→	₹6,000
6-month churn (micro) Right-sized bundle, one bill	23%	→	12%
Status-chase calls Live tracker, WhatsApp alerts	3.0	→	0.5

+382

activations per quarter

368 → 750

−₹10 lakh

acquisition spend per quarter

₹55.4L → ₹45L

+₹26 lakh

new monthly revenue each quarter

382 × ₹6,900 median bundle

Targets are 12-month. Value per quarter assumes the same 5,000 leads; added revenue valued at the ₹6,900 median bundle (conservative vs ₹12,351 average).

Platform Partner unstop

ROLLOUT AND RISKS

Pilot in four cities in 90 days, then scale language by language

0–3 MONTHS — Pilot

Mumbai, Pune, Nagpur, Lucknow

Hindi, Marathi, English · partner QR codes · weekly funnel review

3–6 MONTHS — Expand

All tier 1 cities

+6 languages · WhatsApp front end · partner desk at scale

6–12 MONTHS — Scale

Tier 1 and 2 nationwide

DigiLocker + e-sign · pricing tests · propensity-led field sales

KEY RISKS AND HOW THE DESIGN CONTAINS THEM

AI suggests a wrong product

Rules engine picks products and prices; AI only explains

Fake or misread documents

Registry cross-check, check digit, name match, human review

Partners feel bypassed

Credited for digital sign-ups; complex leads routed to them

Privacy and DPDP Act

Consent per purpose, images deleted, IDs masked, audit log

The ask Approve a 90-day, four-city pilot with partners, tracked weekly on the five metrics.

Demo video · Code on GitHub

Pilot cities: highest-volume tier 1 and tier 2 cities for Hindi and Marathi speakers in smb_leads.csv.

Platform Partner unstop

ANNEXURE · REFERENCE ONLY

Where each case brief deliverable lives, and all links

CASE BRIEF DELIVERABLE → WHERE TO FIND IT

1 Working prototype
Demo video (2:15) and public code repository — Watch the demo · Open the repo

2 Solution architecture diagram
Components and data flows, full page — Slide 5 · full page: p. 13

3 Business case (10 slides or fewer)
Customer, value, pricing, rollout, risks — Slides 2–8 · detail: pp. 10–12

4 1-page data privacy and security note
Choices made against the DPDP Act, 2023 — Slide 5 · full note: p. 14

LINKS

Demo video
youtu.be/fFOwhewZ7_E

Code repository
github.com/ploglabs/udyam-digital

How to run it
README.md: API, web app, tests

Architecture (PDF)
deliverables/Architecture.pdf

Privacy note (PDF)
deliverables/Privacy_Note.pdf

Stack: Next.js · FastAPI · Gemini 2.5 Flash · SQLite. All data synthetic.

Platform Partner unstop

Every major drop reason maps to a step we can automate

Lost at stage	Top reasons owners gave	What Udyam Digital does
Lead 1,594 lost	Unreachable after 3 attempts (805) · Not interested on first call (470) · Wrong number (319)	**Owner starts the chat (QR, WhatsApp, web), so the number is verified and intent is real**
Contacted 1,432 lost	No time for a visit (531) · Locked in with provider (325) · Visit not scheduled (296)	**24×7 self-serve advisor in 10 languages: no visit needed**
Needs assessed 1,113 lost	Price too high (414) · Cheaper competitor (207) · Quote too complex (206) · Too slow (179)	**Instant quote, one monthly price, add-ons the owner can switch on and off**
Quote accepted 273 lost	Missing GST/Udyam doc (94) · Paperwork too long (79) · Details didn't match (47)	**Smart KYC: registry prefill, document AI, clear fix-it messages**
KYC complete 220 lost	Install delayed >21 days (98) · Cancelled before install (53) · Site not feasible (41)	**Feasibility checked before payment; live tracker with an ETA**

Source: smb_leads.csv drop_reason by last stage reached (top reasons shown).

Four bundles, each with one monthly price set by the rules engine

Dukaan Starter

Kirana, salon, pharmacy · 3 staff

₹2,196 /mo

- Biz Fibre 100
- 2 business SIMs
- Cloud POS & billing
- Add CCTV: ₹199 per camera

12 months · setup ₹2,500

Self-serve digital

Clinic & Office Pro

Clinic, CA firm · 8 staff

₹4,592 /mo

- Biz Fibre 300
- 2 priority SIMs
- Email on own domain × 8
- Cloud backup 1 TB
- Endpoint security × 4

12 months · no setup fee

Self-serve digital

Multi-site One

Restaurant/retail · 3 outlets

₹28,045 /mo

- Fibre 300 + 4G/5G failover per site
- Managed SD-WAN
- POS × 3, CCTV × 6
- Managed firewall

24 months · setup ₹49,800

Digital + partner survey

Factory Connect

Small unit · 30 workers

₹22,617 /mo

- Leased line 20 Mbps, 99.5% SLA
- Machine IoT sensors × 6
- 10 SIMs, email × 30
- Managed firewall

24 months · setup ₹32,000

Assisted (feasibility)

Prices from product_catalogue.csv list prices, excluding 18% GST. Bundles generated by the prototype's rules engine for a typical profile.

Smart KYC verifies a business in minutes and closes a compliance leak

Consent first (DPDP Act)

Purpose-wise consent, what we collect and never collect, retention, withdraw anytime

Registry lookup

GSTIN and Udyam checked live: status, legal name, address

Document AI

Gemini vision reads 4 documents; fixes OCR slips only if the check digit agrees

15+ automatic checks

Check digit, active status, names and addresses across documents, 90-day bill, ID expiry

Instant verdict

Approve, re-upload or reject, with a reason the owner understands

12 applicants: 5 approved · 1 re-upload · 6 rejected

- Cancelled GSTIN + blank Udyam number (K01)
- Udyam certificate missing (K05)
- One GSTIN digit changed: check digit fails (K06)
- Edited legal name on GST certificate (K07)
- Address proof 257 days old (K11)
- Blurred certificate: ask to re-upload (K04)

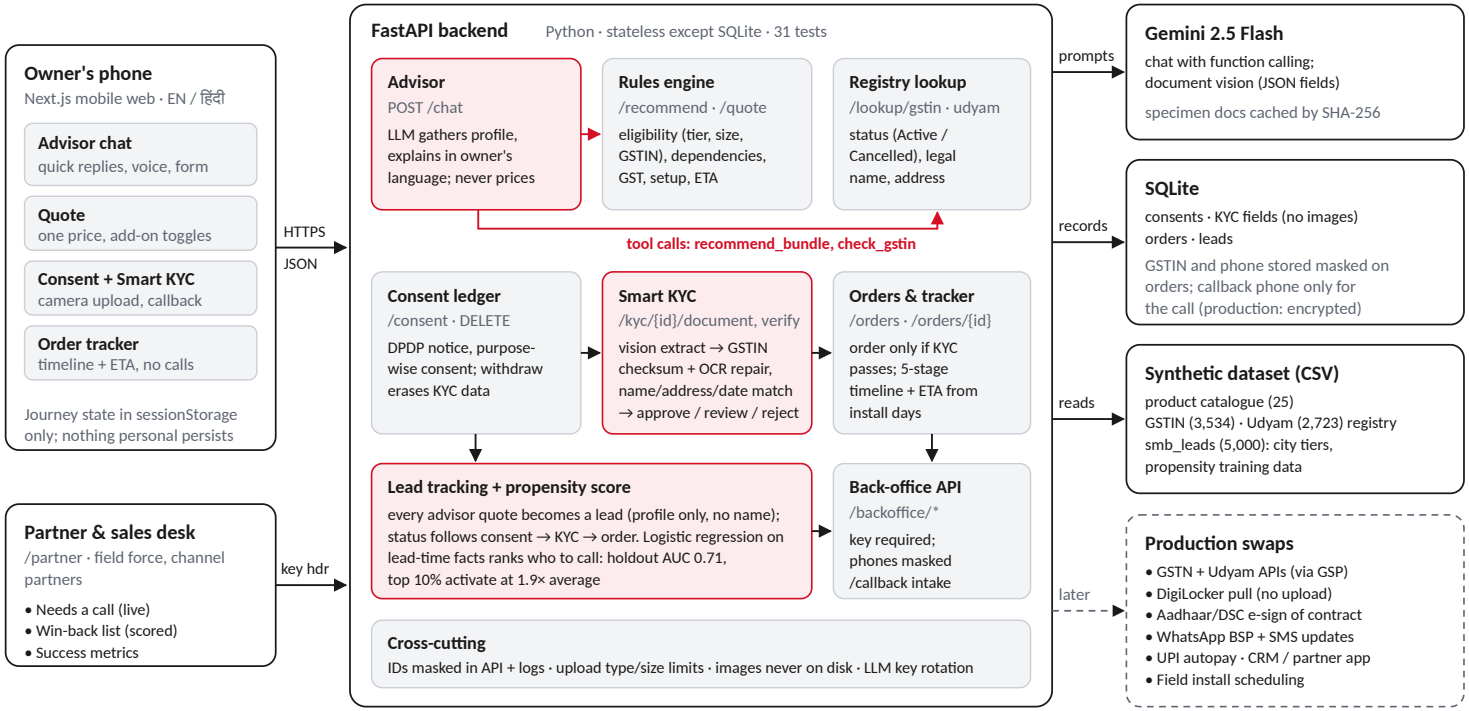
22 customers activated last quarter had a cancelled or suspended GSTIN. A live registry check blocks this at sign-up.

Results from running the prototype on the 12 specimen applicants (K01–K12) and on mock_gstin_registry.csv.

Solution architecture

Udyam Digital · MSME acquisition and onboarding for NorthStar · Airtel SYNC, Enterprise Track A

Solid boxes: built and running in the prototype
Dashed boxes: production integrations



Data flow through one journey

1. **Advise.** The owner chats (or fills the short form). Gemini asks a few questions and must call `recommend_bundle`; the rules engine picks eligible products and prices, so the LLM can explain but never invent a price. A lead row (profile only) is created.
2. **Quote.** `/quote` recomputes the monthly total, 18% GST, setup, contract and activation ETA live as add-ons are toggled; dependency errors block invalid carts.
3. **Consent.** The DPDP notice lists purposes, data collected and not collected, and retention. Required purposes are recorded before any document can be uploaded.
4. **Smart KYC.** Each upload is read by the vision model, the image is dropped, and only fields are kept. `verify` checks GSTIN format and check digit (with safe OCR repair), registry status and names, Udyam, signatory ID validity and bill age (≤ 90 days).
5. **Order and track.** An order needs KYC that is not rejected and an active GSTIN for GST-only products. The tracker shows 5 stages with expected dates, replacing status-chase calls.
6. **Hand-off.** Anything not closed (quote but no KYC, failed KYC, callback request) appears on the partner desk, ranked by propensity, with the next action.

Why it is built this way

- **LLM talks, rules decide.** Prices, eligibility and KYC verdicts are deterministic and unit-tested; the form fallback works if the LLM is down.
- **Minimal data.** No Aadhaar, PAN or bank details asked for; images never stored; identifiers masked in every response and log line.
- **Partners add, not compete.** The digital flow handles the easy 80%; partners get a ranked, reasoned call list instead of cold leads, and can send the WhatsApp advisor link to leads who never had time for a visit.
- **Cheap to run.** One small API, one database, pay-per-call LLM (4 calls per KYC). Scales out by moving SQLite to Postgres and the session store to Redis.

Measured on

12-month targets: lead-to-activation 7.4% → 15% · days to activate 17.6 → 5 · cost per activation ₹15,053 → ₹6,000 · micro-retail 6-month churn 23% → 12% · status-chase calls 3.0 → 0.5.

All businesses, people and identifiers are synthetic (hackathon dataset). The case's fictional operator, Accel Telecom, is branded NorthStar in the prototype.

Data privacy and security note

Udyam Digital · MSME onboarding for NorthStar · Airtel SYNC, Enterprise Track A

Designed against the DPDP Act, 2023

Principle: collect only what each step needs, only after consent, keep it only as long as needed, and never show or log a full identifier. Every point below describes how the working prototype behaves; items marked **Production** are what we would add before launch.

1 · Consent before any KYC data

- A notice is shown before any document upload: purposes, data collected, data *not* collected, retention, grievance contact.
- Consent is per purpose. *KYC verification* and *service activation* are required. *Marketing* is optional and off by default.
- Each consent gets an ID and a timestamp (audit trail). Withdrawing it from the KYC screen erases that application's KYC data.

2 · Purpose limitation

- KYC data is used only to verify the business and activate the services ordered.
- The GenAI advisor needs no personal data. It is instructed never to ask for Aadhaar, PAN, bank details or documents in chat.
- Marketing messages only with the separate, optional consent.

3 · Data minimisation: what we collect, and why

Step	Collected	Why it is needed	Never collected
Advisor	Business type, city, staff, sites, needs (no names or IDs)	Pick an eligible bundle and price	Aadhaar number, bank or card details, personal photos, date of birth
KYC	GSTIN, Udyam no., business name and address, signatory name and ID validity, bill date	Verify the business is real, active and at that address	
Order	Mobile number	Order updates and e-sign link	
Callback (optional)	Mobile number, only if the owner taps "Talk to a person"	So a local partner can call back	

4 · Storage and retention

- **Document images are never stored.** They are processed in memory, the fields are extracted, and the image is discarded.
- Only the extracted fields and the verdict (approve / review / reject with reasons) are kept, linked to the consent ID.
- Retention: for the contract term, plus any period telecom KYC rules require. The prototype states contract term + 8 years; **Production:** confirm the exact period with legal against DoT rules. Then delete automatically.

5 · Masking everywhere

- Identifiers are masked in the UI, in API responses and in the order record: GSTIN `32TT********1ZW`, Udyam `UDYAM-KL-*******801`, phone `98******10`.
- Logs record only the KYC application ID, document type and how it was read, never field values.
- Journey state lives in the browser's session storage and is gone when the tab closes.
- The partner desk is behind an access key and shows leads as a business profile (segment, size, city tier) with a masked callback number, never names or IDs.

6 · Use of AI (Gemini)

- Gemini receives only the uploaded document image (to read it) and the chat text. No stored customer records are sent.
- AI never makes the KYC decision alone. Deterministic rules decide: GSTIN check digit, registry status, cross-document matching. AI output that fails the check digit is rejected.
- **Production:** move from the free API tier to an enterprise tier with no training on customer data and India data residency.

7 · Security controls

- Uploads limited to PNG, JPEG, WEBP or PDF up to 8 MB. Typed validation of API inputs (e.g. Indian mobile number format).
- API keys kept in an environment file, never in code or the repository.
- **Production:** TLS everywhere, encrypted database and backups, role-based access and audit logs for the partner/back-office view, rate limiting, secrets in a vault, and annual security testing.

8 · Owner rights

- Access: the order tracker shows the business details, services and status we hold.
- Correction: re-upload any document. Erasure: withdraw consent before activation; after activation, on request once legal retention ends.
- A named grievance contact is shown on the consent screen.

9 · Breach response

- Because images aren't stored and identifiers are masked, a breach exposes far less.
- Incident runbook: contain, assess, then notify the Data Protection Board and affected owners as the DPDP Rules require. Then review and fix.

All businesses, people and identifiers in the prototype are synthetic (hackathon dataset). The case's fictional operator, Accel Telecom, is branded NorthStar in the prototype.